Lab 1

An assignment for class **MET CS579** Database Management at Boston University due on **February 9, 2023**, instructed by **Jae Lee**. Students Jin Yu Zhang and Wei Wang submitted on **February 8, 2023**.

Assumptions

Description for things not included in the requirements:

- Customer is not an entity since it doesn't have any attribute, and we assume that each customer has only one headquarter by the definition of a headquarter. Therefore, customer and **HEADQUARTER** are equivalent here
- Each customer has one or more branch stores. We assume that there is no limit for the number of **Branch_Store** for a **HEADQUARTER**
- According to the requirements, we should keep the information about the retail store who issued the order for an order. However, since there is already a relationship between **ORDER** and **HEADQUARTER**, we will not put this as an attribute in **ORDER**
- According to the requirements, we should keep the information about the employee(s) of the Boston Shoes who handle the order for an order. However, since there is already a relationship between **ORDER** and **EMPLOYEE**, we will not put this as an attribute in **ORDER**. Also the requirement says an employee handles at most four order, we assume an employee is allowed to handle no orders
- There are not limits for how much suborders a **BRANCH_STORE** can be shipped to, and how many suborders can a **SHOES_TYPE** hold. There are hence form with **N** (many) constraint in this assignment
- We assume an **EMPLOYEE** come with only one classification

Questions

Known issues:

- Is sequence number a derived value? If it is, how do we define the partial key for **SUBORDER** and **BRANCH_STORE**?
- 5-2, 5-4, and 6-2. Should we store them as attributes or relationship types are enough?

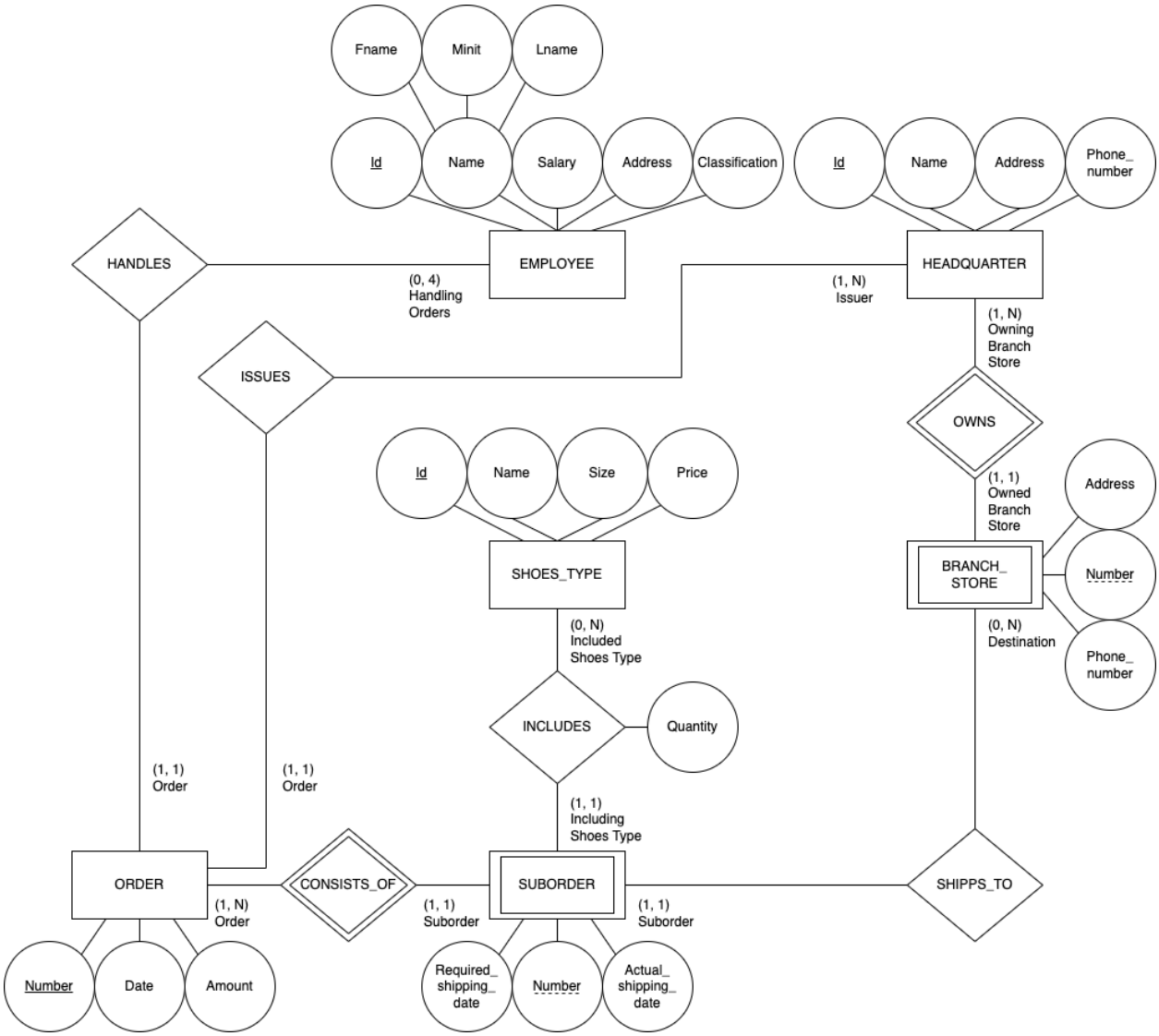
Entity Types

Name	Description	Attribute			
		Name	Description	Multivalued	Derived
EMPLOYEE	Employees works for Boston Shoes.	Id {K} Name Fname Minit Lname Salary Address Classification		N N N N N	N N N N N
SHOES_TYPE		Id {K} Name Size Price		N N N N	N N N N
ORDER		Number {K} Date Amount		N N N	N N N
SUBORDER		Number {WK} Required_shipping_date Actual_shipping_date	Number is a sequence number starting with 1 and is only unique among those that belong to a particular order.	N N N	N N N
HEADQUARTER	The customers of Boston Shoes such as Walmart, Target, and Kohls.	Id {K} Name Address Phone_number		N N N N	N N N N
BRANCH_STORE		Number {WK} Address Phone_number	Number is a sequence number	N N N	N N N

			starting with 1 and is only unique among those that belong to the same retail store.		

Relationship Types

Name	Description	Participating Entity Types		Attribute
		Name	Structural Constraint	
OWNS	Each customer (headquarter) has one or more branch stores.	HEADQUARTER BRANCH_STORE	(1, N) (1, 1)	None
ISSUES	An order is issued by a retail store headquarter.	HEADQUARTER ORDER	(1, N) (1, 1)	None
HANDLES	Boston Shoes employees handle the order.	EMPLOYEE ORDER	(0, 4) (1, 1)	None
CONSISTS_OF	An order consists of one or more suborders.	SUBORDER ORDER	(1, 1) (1, N)	None
SHIPS_TO	A suborder is shipped to exactly one branch store.	SUBORDER BRANCH_STORE	(1, 1) (0, N)	None
INCLUDES	A suborder includes one or more shoe types.	SUBORDER SHOES_TYPE	(1, N) (0, N)	Quantity


Fname
Minit
Lname
Id
Name
Salary
Address
Classification
Id
Name
Address
Phone_ number
HANDLES
EMPLOYEE
(0, 4) Handling Orders
(1, N) Issuer
HEADQUARTER
(1, N) Owning Branch Store
ISSUES
OWNS
Id
Name
Size
Price
(1, 1) Owned Branch Store
Address
SHOES_TYPE
BRANCH_ STORE
Number
(0, N) Included Shoes Type
(0, N) Destination
Phone_ number
INCLUDES
Quantity
(1, 1) Order
(1, 1) Order
(1, 1) Including Shoes Type
ORDER
CONSISTS_OF
SUBORDER
SHIPPS_TO
(1, N) Order
(1, 1) Suborder
(1, 1) Suborder
Number
Date
Amount
Required_ shipping_ date
Number
Actual_ shipping_ date